

SIMATS ENGINEERING

ASSESSMENT TOOL 2

Course Name: Deep Learning
Course Code: MLA0408
Name / Registration No: Maniyar Mohammad Rehan (192425391)

Question 1 (50 Marks)

A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Answer

The observed training and testing errors are summarised below before proceeding with the analysis:

Metric	Value
Training MSE	0.85
Testing MSE	5.20
Ratio (Test / Train)	$\approx 6.1\times$

1. Error Identification

The training MSE (0.85) is significantly lower than the testing MSE (5.20) — the test error is over six times the training error. This large gap is a classic symptom of overfitting, where the model performs very well on data it has already seen but poorly on new, unseen data. The specific errors identified are:

- **High variance error:** the model's performance changes drastically between training and testing sets, showing it has memorised training examples rather than learning generalisable patterns.
- **Poor generalisation error:** the large test error indicates the learned weights do not capture the true underlying relationship between input features (e.g., attendance, assignment scores, study hours) and the final student score.
- **Train–test distribution mismatch:** if the test set differs in distribution (different student demographics, score ranges, or feature coverage not seen in training), error can spike even without classic overfitting.
- **Optimistic training performance:** a very low training MSE of 0.85 suggests the model may be fitting noise or outliers specific to the training samples rather than the true trend.
- **Possible data leakage check needed:** if training MSE were unrealistically close to zero this could indicate leakage of test information into training; here it should still be verified that no overlapping records exist between the two sets.

2. Error Analysis & Reasoning

Several factors, individually or in combination, could explain this gap between training and testing performance:

- **Model complexity too high:** an overly deep or wide network (too many layers/neurons) relative to the size and complexity of the dataset can fit noise in the training data, producing a low training error but a high test error.
- **Insufficient training data:** a small dataset gives the network too few examples to learn general patterns, so it defaults to memorising the samples it has seen instead of learning the underlying trend.
- **Lack of regularisation:** without techniques such as L1/L2 weight penalties or dropout, the network is free to assign large weights to noise-specific features, increasing variance.
- **Training for too many epochs:** prolonged training without early stopping allows the network to keep reducing training loss well past the point of optimal generalisation, a pattern typically visible as a widening gap between training and validation loss curves.
- **Improper or non-representative data splitting:** if the train/test split was not randomised or stratified properly, the test set may contain patterns or score ranges not represented in the training data.
- **Noisy or unrepresentative test data:** outliers, mislabelled records, or missing values in the test set can inflate the MSE independent of the model's actual generalisation ability.
- **Feature scaling/preprocessing inconsistency:** if normalisation statistics (mean/variance) were computed only on the training data but applied incorrectly to the test set, or if features were not scaled at all, predictions on test data can be distorted.
- **Class/score imbalance:** if certain score ranges are under-represented in training, the model will predict poorly for those ranges when they appear more frequently in the test set.
- **Inadequate learning rate or optimisation settings:** a learning rate that is too high or too low, or a poorly tuned optimiser, can cause the model to settle into a sharp minimum that fits training data well but generalises poorly.
- **Random weight initialisation and lack of cross-validation:** relying on a single train/test split without cross-validation means the reported gap could partly reflect an unlucky split rather than only overfitting.

3. Correction Strategy

The following corrective measures are recommended, each targeting one or more of the causes identified above:

- **Add regularisation:** introduce L2 (weight decay) or L1 regularisation to the loss function, and/or use dropout layers to prevent co-adaptation of neurons and reduce overfitting.
- **Reduce model complexity:** decrease the number of hidden layers/neurons, or use pruning, so that model capacity better matches the size and complexity of the dataset.
- **Increase and diversify training data:** collect more labelled samples or apply data augmentation/synthetic sampling to expose the model to a wider variety of patterns.

- **Use early stopping:** monitor validation loss during training and stop once it begins to rise, even if training loss continues to fall, to prevent the model from over-fitting later epochs.
- **Apply k-fold cross-validation:** this gives a more reliable estimate of generalisation performance, reduces sensitivity to a single train/test split, and helps tune hyperparameters robustly.
- **Ensure a proper train/validation/test split:** use stratified, randomised splitting so all sets reflect the same underlying data distribution and score ranges.
- **Normalise/standardise features consistently:** fit scaling parameters only on the training set and apply the same transformation to the validation/test sets to avoid distribution mismatch.
- **Tune hyperparameters carefully:** adjust learning rate, batch size, and number of epochs using the validation set (not the test set), keeping the test set purely for final, unbiased evaluation.
- **Use batch normalisation:** stabilises and speeds up training, which can also have a mild regularising effect and improve generalisation.
- **Ensemble methods:** combining predictions from multiple models (e.g., bagging) can reduce variance and improve robustness on unseen data.
- **Simplify or engineer features:** remove noisy/irrelevant input features and engineer more informative ones so the model focuses on genuinely predictive signals rather than incidental correlations.

4. Interpretation of Results

The large gap between training and testing MSE means the model, in its current state, cannot be trusted to make accurate score predictions for new students — it has effectively memorised the training set rather than learning the true relationship between features and scores. Deploying this model as-is would likely produce unreliable, high-error predictions in real use.

This case illustrates the classic bias–variance trade-off: the model currently has low bias (it fits the training data very well) but high variance (it does not generalise to new data). The goal after applying the corrective measures above — particularly regularisation, early stopping, cross-validation, and more/better data — is to bring the training and testing MSE closer together at a reasonably low value. A well-generalised model would typically show training and testing errors of a similar, modest magnitude, with only a small, expected gap.

Continuous monitoring of training vs. validation loss curves in future training runs is recommended, since a widening gap early in training is the clearest practical signal to intervene before the model is finalised and deployed.

Conclusion

The difference between the training MSE (0.85) and testing MSE (5.20) is primarily attributable to overfitting caused by high model complexity, insufficient regularisation, and possibly limited or imbalanced training data. Applying regularisation techniques, early stopping, cross-validation, proper data splitting, and consistent preprocessing should substantially reduce this gap and produce a neural network that generalises reliably to new student data.

Code of Conduct

I Maniyar Mohammad Rehan (192425391) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Mohammad Rehan